

Evaluating Duty-Cycle Enforcement in Distance-Vector LoRa Mesh Networks: A Cross-Layer ns-3 Study

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Abstract—Multi-hop LoRa (LoRa mesh) can extend coverage and resilience beyond LoRaWAN star topologies, but its end-to-end performance degrades when duty-cycle (DC) constraints are enforced and traffic is no longer sporadic. However, many LoRa mesh evaluations either do not enforce DC explicitly or do not record metrics that connect DC enforcement with control-plane degradation. This paper evaluates the impact of DC enforcement in a cross-layer LoRa mesh stack with proactive Distance-Vector (DV) routing that advertises routes through a compact scalar score and a CSMA/CA MAC based on Channel Activity Detection (CAD), with explicit 1% DC applied to both data and beacons. We evaluate this stack under any-to-any traffic in a 1 km² area for network sizes $N = 9, 16, 25, 36, 49$, and 64 randomly located nodes, under low and medium loads, and 10 seeds per configuration (240 runs). Results show that enabling DC systematically reduces the end-to-end delivery ratio (DR): for $N = 25$ it drops from 0.958 to 0.438 under low load and from 0.159 to 0.058 under medium load, while for $N = 64$ the relative impact is smaller (from 0.399 to 0.314 and from 0.051 to 0.039) due to a transition to a contention-dominated regime. Control-plane logging is consistent with this mechanism: DC reduces beacon health (H_b), which is accompanied by DV route expirations and more no-route events, linking DC enforcement to end-to-end losses.

Index Terms—LoRa mesh, Distance-Vector routing, Duty-cycle constraints, CSMA/CA, CAD, ns-3, Multi-hop IoT.

I. INTRODUCTION

LoRa and LoRaWAN have enabled long-range, low-power IoT deployments, but their star-of-stars architecture limits connectivity in scenarios without gateway coverage, with sparse topologies, or where multi-hop resilience is required [1]. LoRa mesh networks address this limitation by introducing multi-hop forwarding, at the cost of increased contention, collisions, and control overhead on a shared channel [1], [2].

A frequently underestimated factor in LoRa mesh is the regulatory duty-cycle (DC) constraint in ISM bands (e.g., EU868 [2]–[4]), which limits the fraction of airtime that each device may occupy and imposes mandatory silence windows that affect both the data plane and the control plane (beacons and data packets). In EU868-oriented deployments, commonly used sub-bands for LoRa/LoRaWAN operation are subject to low DC limits (often 1%), making DC a practical bottleneck even at moderate offered loads [3], [4]. In parallel, several works

explore proactive routing, spreading factor (SF) adaptation, and hybrid approaches to extend connectivity beyond star topologies [5]–[7]; in particular, [8] presents a minimalist DV protocol for LoRa mesh with SF selection aimed at minimizing airtime, which we use as a methodological reference.

However, despite the existence of LoRa mesh proposals, many studies either do not enforce DC explicitly in simulation or treat it only as a scenario parameter, without recording its effect on control-plane freshness, route expirations, or channel admission [9], [10]. Specifically, [9] evaluates routing under DC constraints but reports only end-to-end indicators, leaving the link between DC enforcement and routing-table dynamics implicit. This creates a methodological gap: without explicit DC enforcement and control-plane telemetry, it is difficult to attribute performance degradation to contention, DC temporal limitations, or route instability.

In this work, we quantify through simulation the impact of explicitly enforcing a 1% DC (representative of common EU868 sub-band limits) and its interaction with the DV control plane and the data plane, using a reproducible campaign that combines end-to-end metrics (DR and latency) with detailed control-plane and channel-admission records.

A. Contributions

This work provides new evidence on how regulatory duty-cycle (DC) constraints shape LoRa mesh performance:

- **Quantified impact of enforcing 1% DC:** We evaluate the impact of a 1% DC constraint under varying network sizes and traffic loads, showing that DC can cause substantial end-to-end delivery ratio (DR) reductions at small/medium N , while its relative impact attenuates in contention-limited regimes at high density/load.
- **Cross-layer measurement methodology:** We jointly log control-plane health (H_b), DV expirations (E_{DV}), no-route events (D_{NR}), source admission (R_{tx0}), and DC/CAD event counts (B_d , B_c) together with end-to-end metrics. While the qualitative DC $\rightarrow$ DR effect is predictable, prior LoRa-mesh studies under DC [9] report only end-to-end outcomes. We expose the explicit causal chain (DC $\rightarrow$

H_b loss $\rightarrow$ route expirations $\rightarrow$ no-route drops $\rightarrow$ DR), enabling mechanistic rather than correlational attribution.

- **Latency interpretation under low delivery:** We show that computing latency only over delivered packets introduces survivorship bias when DR and R_{tx0} are low, and we discuss how to interpret mean delay and D_{95} in conjunction with DR , R_{tx0} and N_{del} in such regimes.

II. SYSTEM MODEL AND PROTOCOL DESIGN

A. Stack and ns-3 implementation

The evaluated system is a multi-hop LoRa mesh network implemented in ns-3. Fig. 1 summarizes the stack: (i) a proactive DV control plane based on beacons, (ii) a multi-hop unicast data plane, and (iii) a MAC layer with CAD-based CSMA/CA and explicit DC accounting (1%). The simulator exports control-plane logs (scheduled/sent/blocked beacons and DV expirations) and end-to-end metrics (DR and latency, conditioned on successful deliveries). In Fig. 1, *Application* generates low/medium offered load and timestamps packets; *Routing* maintains DV state and selects next hops from beacon-learned routes; *MAC* performs CAD, applies backoff on busy detections, and enforces DC via time-on-air accounting and mandatory silent intervals (for both beacons and data); *Device* provides L2 encapsulation and receive filtering for unicast/forwarding with TTL, and *PHY/Channel* models LoRa settings plus propagation and interference/capture, determining reception outcomes that drive DR and latency.

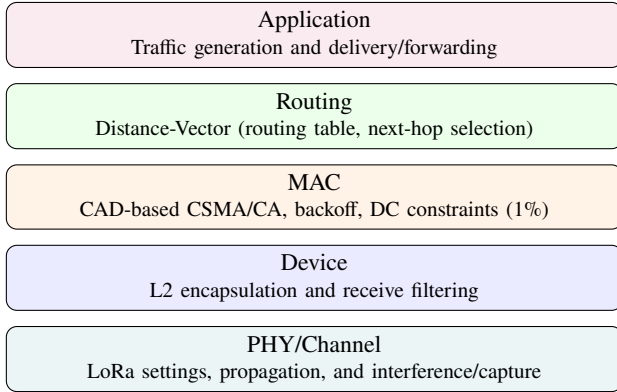


Fig. 1. Evaluated stack: proactive DV (control plane), multi-hop unicast (data plane), and CAD-based CSMA/CA with DC constraints.

B. PHY/channel and parameters

We consider a single LoRa channel at 868 MHz (BW 125 kHz, CR 4/5). Spreading-factor (SF) selection is hop-by-hop: data packets choose an SF per neighbor (learned empirically during setup), and beacons use a probabilistic SF policy inherited from [8], avoiding ADR decisions based on SNR/RSSI along the active data path. We assume a single-demodulator LoRa receiver (one reception at a time per node), without concurrent multi-SF demodulation. For methodological comparability with [8], we adopt a fixed capture model (6 dB threshold) and treat SFs as quasi-orthogonal (collisions are

App Layer: Data Header (9 Bytes)					Payload (Variable Bytes)		
Src ID (2 B)	Dst ID (2 B)	NextHop ID (2 B)	Type+TTL (1 B)	Seq16 (2 B)			
App Layer: Beacon Header (5 Bytes)			DV Payload Entries (N × 3 Bytes)				
Src ID (2 B)	Dst ID (0xFFFF) (2 B)	Type+TTL (1 B)	Dest 1 (2 B)	Score 1 (1 B)	...	Dest N (2 B)	Score N (1 B)

Fig. 2. On-air format: data header (top) and beacon (bottom).

destructive only for same-SF packets). Propagation uses LogDistancePropagationLossModel ($n = 2.08$, $d_0 = 40$ m, reference loss $L(d_0) = 127.41$ dB) plus i.i.d. log-normal shadowing in dB, $\mathcal{N}(0, 3.57^2)$ dB, via RandomPropagationLossModel, no fast fading is enabled. Delay uses ConstantSpeedPropagationDelayModel. All PHY/channel parameters above match the baseline configuration used for all runs in this campaign.

C. On-air packet format

To avoid ambiguities between internal metadata and fields transmitted over the radio, we specify the on-air format used in the campaign. As shown in Fig. 2, data packets are multi-hop unicast and include end-to-end identifiers (`src`, `dst`) and the next hop for the current hop (`via`), plus a `type/TTL` field and a `seq16` identifier for end-to-end (E2E) deduplication. Beacons are 1-hop broadcasts and include `src`, `dst=0xFFFF` and a `type/rp_counter` field (6 bits) that is incremented only when the beacon is actually transmitted on the air (not when it is merely scheduled). Each DV entry is advertised compactly as `destination` (2 B) + `score` (1 B).

D. DV routing and composite metric

Routing is proactive: nodes periodically broadcast *beacons* and update their DV table upon receiving neighbor advertisements. The control plane is *score-only*: each advertised entry carries `destination+score` (1 byte, range 0...100). Internally, we use a normalized cost $C \in [0, 1]$ (lower is better), mapped to the advertised score as

$$score = \text{round}(100 \cdot (1 - C)), \quad (1)$$

with $score = 0$ reserved for poison/unreachable semantics. Accordingly, the path cost advertised by neighbor j towards destination d is

$$C_{j \rightarrow d}^{\text{path}} = 1 - \frac{score_{j \rightarrow d}}{100}. \quad (2)$$

Let C_{ij}^{link} be the local link cost from i to neighbor j . The candidate cost at i to reach d via j is combined through a bottleneck rule:

$$C_{i \rightarrow d}^{\text{cand}} = \max(C_{ij}^{\text{link}}, C_{j \rightarrow d}^{\text{path}}). \quad (3)$$

The composite link cost is

$$C_{ij}^{\text{link}} = \alpha \cdot \text{ToA}_{ij}^{\text{norm}} + \beta \cdot h_{ij}^{\text{norm}} + \gamma \cdot \Psi(b_j), \quad (4)$$

where $\text{ToA}_{ij}^{\text{norm}} = \frac{\text{ToA}_{ij} - \text{ToA}_{\min}}{\text{ToA}_{\max} - \text{ToA}_{\min}}$, with $\text{ToA}_{\min} = \text{ToA}(\text{SF7}) = 143360 \mu\text{s}$ and $\text{ToA}_{\max} = \text{ToA}(\text{SF12}) =$

3293184 μ s computed using the standard LoRa time-on-air formula (BW=125 kHz, CR=4/5) for the fixed frame format used in this campaign. The hop term is $h_{ij}^{\text{norm}} = \min(h_{ij}/h_{\max}, 1)$ with $h_{\max} = 10$. The energy penalty is $\Psi(b_j) = 1 - b_j^p$, with $p = 2$ and $b_j = E_j^{\text{rem}}/E_j^{\text{max}} \in [0, 1]$; battery is not advertised on-air, and affects routing only through the propagated score.

In this campaign we fix $\alpha = 0.40$, $\beta = 0.30$, and $\gamma = 0.30$ across all runs to avoid scenario-specific tuning. Route selection prioritizes lower cost (equivalently, higher score); ties are broken by preferring lower SF. To reduce oscillations, hysteresis is applied: changing the next hop requires exceeding a minimum score delta.

E. CSMA/CAD MAC with DC enforcement

At the MAC layer we enable CAD-based CSMA/CA: before transmitting, a node performs CAD and applies slot-based backoff if the channel is busy, following CSMA/CAD mechanisms studied for LoRa [11], [12]. The system does not implement ARQ/acknowledgments: losses due to collisions, DC blocking or queue drops are not recovered through explicit retransmissions.

When enabled, DC enforcement limits the cumulative transmission time per node to 1% over a sliding window of 3600 s (36 s of airtime per hour), consistent with typical EU868 regulation [3]. The MAC accounts for per-packet ToA and blocks transmissions that exceed the budget. This mechanism affects both the data plane and beacons; therefore we instrument the ratio `beacon_tx_sent/beacon_scheduled` and DC-induced blocking events in the control plane.

TABLE I
BASELINE CAMPAIGN PARAMETERS (CONFIGURATION FROZEN FOR THIS PAPER).

Parameter	Value
Area / deployment	1 km ² , random static nodes
Network sizes N	{9, 16, 25, 36, 49, 64}
Traffic	any-to-any; 20 B payload
Loads	low: 1 pkt/100 s/node; medium: 1 pkt/10 s/node
Application jitter	uniform, up to 3 s per packet
Band/PHY	868 MHz, BW 125 kHz, CR 4/5, single channel
MAC	CSMA/CA with CAD; min backoff 4 slots, step 2
DC	off / on (1%); 3600 s window
TX power	20 dBm
Interference	6 dB capture, quasi-orthogonal SFs [8]
beacons	$T_b = 120$ s; $T_{\text{rt}} = 7 T_b$
Data window	300–3900 s; end at 4500 s
DR and latency	last 600 s excluded; 600 s drain period
E2E deduplication	key (src,dst,seq16), 600 s window
Seeds	10 per configuration

III. METHODOLOGY

A. Simulation environment and comparability

Experiments are carried out in ns-3 using our own LoRa mesh implementation. We take Pueyo-Centelles et al. [8] as methodological reference for parameters, assumptions and scenario comparability. In this work the focus is to quantify the impact of 1% DC enforcement and measure its effect on the control and data planes; a bit-level implementation and direct comparison against the baseline is left as future work.

B. Scenarios

We consider static random deployments in a 1 km² area with $N \in \{9, 16, 25, 36, 49, 64\}$ nodes. All nodes operate on a single LoRa channel at 868 MHz with BW 125 kHz and CR 4/5. TX power is fixed at 20 dBm. To preserve comparability with [8], we use a capture/interference model consistent with that baseline.

C. DV routing and MAC configuration

Nodes run a proactive DV protocol that disseminates routes via 1-hop broadcast beacons. The steady-state beacon interval is set to $T_b = 120$ s and the route timeout to $T_{\text{rt}} = 7 T_b = 840$ s. Metrics are reported over the evaluation window defined by the traffic (see below).

At the MAC layer we enable CSMA/CA supported by CAD, with a minimum backoff of 4 slots and a backoff window increment of 2 slots. No ARQ/acknowledgment is implemented; therefore, losses due to collisions, DC blocking or congestion are not recovered through explicit retransmissions.

D. DC enforcement

DC enforcement follows the MAC described in Sec. II-E. transmissions are subject to explicit time-on-air accounting and mandatory silent intervals under a 1% DC, applied uniformly to data and DV beacons. The simulator logs DC-related outcomes (e.g., blocked/postponed transmissions) together with CAD/busy events to support the admission and control-plane analyses.

E. Any-to-any traffic

Traffic is any-to-any. Each node generates packets with a 20 B payload and, at each transmission, chooses the destination uniformly at random among all node IDs except itself. Two presets are defined:

- Low: 1 packet every 100 s per node (0.01 pkt/s/node).
- Medium: 1 packet every 10 s per node (0.1 pkt/s/node).

To avoid artificial synchronization, a random jitter of up to 3 s is applied per packet. Traffic is active between $t = 300$ s and $t = 3900$ s; the simulation ends at $t = 4500$ s. We stop generating traffic at $t = 3900$ s and leave 600 s to drain queues.

F. Metrics

We report end-to-end metrics together with control-plane and channel-admission logs.

a) *End-to-end delivery*: The end-to-end *delivery ratio* (DR) is defined as $DR = N_{\text{del}}/N_{\text{gen,elig}}$, where $N_{\text{gen,elig}}$ excludes packets generated during the last 600 s. Packets are identified by the E2E key (src, dst, seq16) and are deduplicated using a 600 s time window to bound memory usage and mitigate sequence number wrap-around effects.

b) *End-to-end latency*: Latency is computed only over delivered packets. For each run, we report (i) the mean end-to-end delay $\bar{D}$ and (ii) the 95th-percentile delay D_{95} (seconds). Being conditioned on successful deliveries, latency must be interpreted jointly with DR and with the sample size N_{del} .

c) *Channel admission*: We report $R_{tx0} \in [0, 1]$, the fraction of generated packets that obtain at least one TX attempt at the source.

d) *Control-plane health, stability and proxies*: We define $H_b = N_{b,tx}/N_{b,sched}$ as the ratio between beacons effectively transmitted and beacons scheduled, and we report DV route expirations E_{DV} (total expiration events over the evaluation window). We denote by D_{NR} the number of data packets dropped due to missing routes (*no-route*). As indirect indicators, we record B_d (DC blocking/postponement events per node) and B_c (busy CAD events per node).

G. Statistical procedure and reproducibility

Each configuration is run with 10 independent seeds; we report the mean and 95% confidence intervals (Student- t) and export a CSV/JSON summary per run.

IV. RESULTS AND DISCUSSION

We report results from the main campaign (240 runs) with $N \in \{9, 16, 25, 36, 49, 64\}$ nodes, low/medium loads, DC off/on (1%) and 10 seeds per configuration. Figs. 3–4, Figs. 5–6, and Table II summarize end-to-end performance together with control-plane and admission logs.

TABLE II

SUMMARY OF RESULTS (MEANS OVER 10 SEEDS) FOR $N \in \{25, 64\}$ UNDER LOW/MEDIUM LOADS AND DC OFF/ON (1%). DR : END-TO-END DELIVERY RATIO; R_{tx0} : FRACTION WITH AT LEAST ONE TX ATTEMPT AT THE SOURCE; H_b : BEACONS SENT/SCHEDULED; E_{DV} : DV EXPIRATIONS; D_{NR} : NO-ROUTE EVENTS.

N	Load	DC	DR	R_{tx0}	H_b	E_{DV}	D_{NR}
25	low	off	0.9583	0.9760	0.9677	0.00	3.10
25	low	on	0.4380	0.4727	0.9110	112.90	10.80
25	medium	off	0.1594	0.1715	0.9476	0.00	6.90
25	medium	on	0.0581	0.0643	0.8898	85.90	115.70
64	low	off	0.3988	0.4296	0.7578	0.00	5.10
64	low	on	0.3139	0.3474	0.7207	359.60	9.30
64	medium	off	0.0512	0.0561	0.7659	0.00	7.70
64	medium	on	0.0392	0.0440	0.7232	128.30	33.50

A. End-to-end delivery and regime transition

Fig. 3 shows DR . Under *low* load, DR remains high for small/medium network sizes and decreases with N due to contention and multi-hop forwarding. Enabling DC reduces DR across the entire range of N , with a stronger relative impact on small/medium networks and a smaller impact on dense networks where the system operates closer to a contention-dominated regime. Under *medium* load, the system enters a high-contention regime earlier, as reflected in low DR even with DC off. Across all configurations, the variability over the 10 seeds remains limited, which indicates that the observed degradation with DC is consistent and driven by the temporal DC mechanism rather than by atypical realizations.

Fig. 4 reports R_{tx0} , the fraction of generated packets that obtain at least one transmission attempt at the source. Low values of R_{tx0} indicate that a significant fraction of the offered traffic does not get admitted to the channel. At high densities (e.g., $N = 64$) under *medium* load, R_{tx0} drops to very low values,

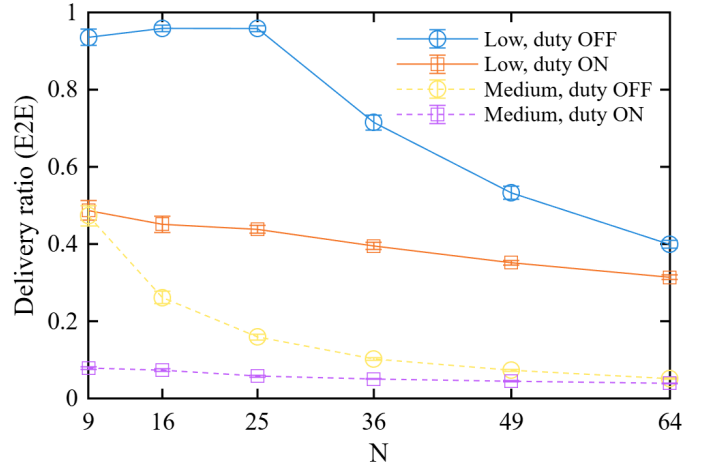


Fig. 3. End-to-end delivery ratio (DR) vs. N with 95% confidence intervals for low/medium loads and DC off/on (1%). Each point corresponds to the mean over 10 seeds.

characterizing a contention-limited regime. In that regime, the relative impact of DC on DR is attenuated because channel admission becomes the dominant limitation.

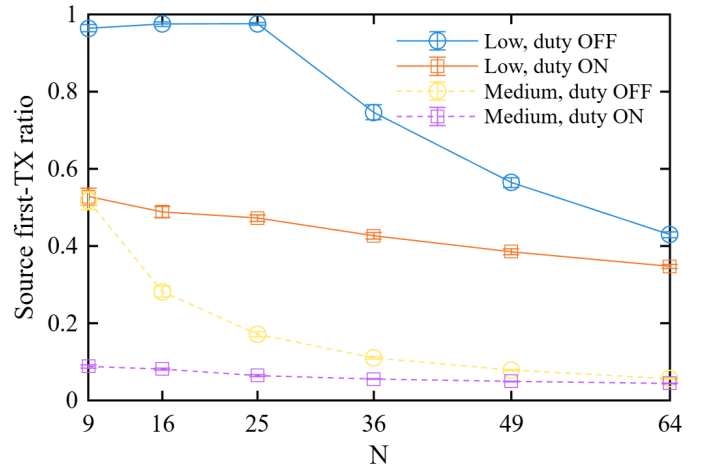


Fig. 4. Fraction of packets that obtain at least one transmission attempt at the source (R_{tx0}) vs. N with 95% confidence intervals for low/medium loads and DC off/on (1%). Low values indicate a contention-dominated admission regime.

B. DV control-plane degradation under DC

Fig. 5(a) reports $H_b = N_{b,tx}/N_{b,sched}$, which quantifies the fraction of beacons effectively transmitted with respect to those scheduled. With DC on, ToA accounting introduces blocking/postponement events that reduce H_b when the time budget is exhausted and/or when the channel is frequently busy. This degradation is reflected in route stability: Fig. 5(b) shows that DV expirations (E_{DV}) emerge under DC on, whereas they remain negligible under DC off in this campaign. At very high densities, E_{DV} may not grow monotonically with N due to the transition to a contention-limited regime: as effective control-plane activity decreases (fewer timely transmissions),

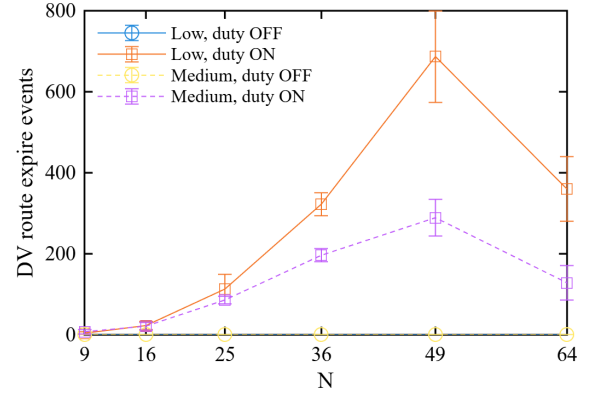
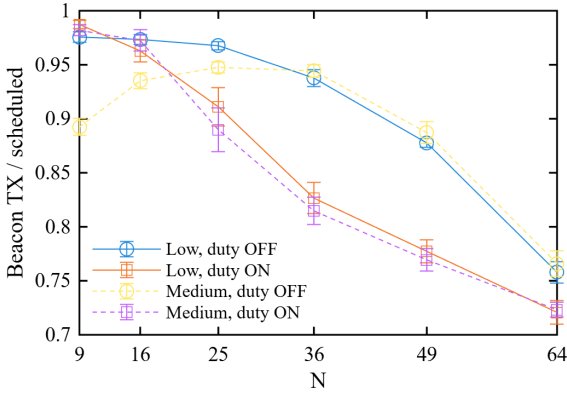


Fig. 5. DV control-plane logging: (a) control-plane health (H_b) and (b) DV route expirations (E_{DV}) vs. N for low/medium loads and DC off/on.

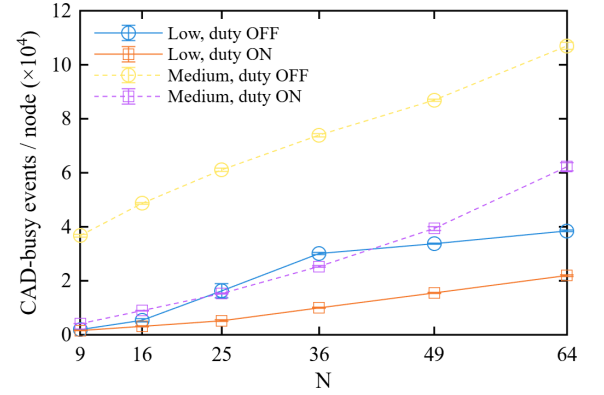
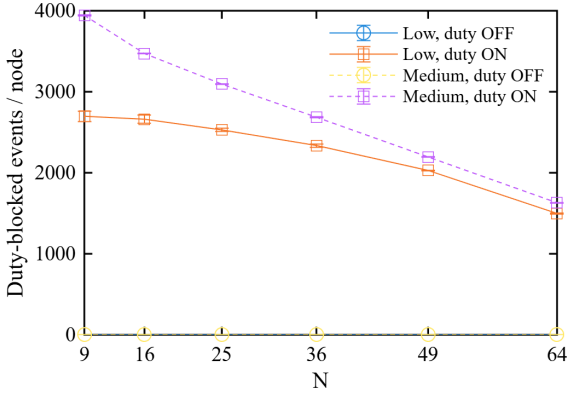


Fig. 6. Indirect indicators of postponement and channel occupancy: (a) DC blocking/postponement events per node (B_d) and (b) busy CAD events per node (B_c), both vs. N with 95% confidence intervals. These counts approximate the intensity of DC-related temporal restrictions and contention (busy channel), respectively.

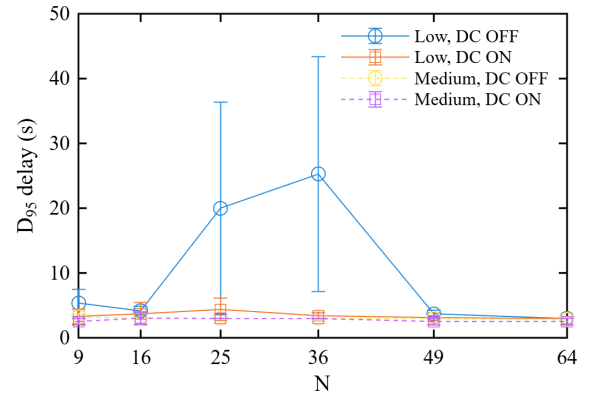
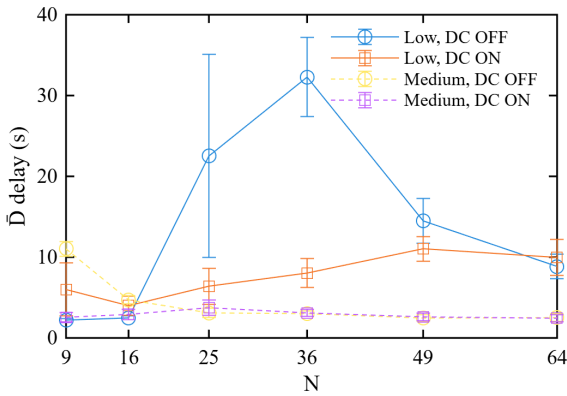


Fig. 7. End-to-end latency ($\bar{D}$ and D_{95}) vs. N under low/medium loads and DC off/on (1%). Each point corresponds to the mean over 10 seeds.

the expiration counter can decrease even while end-to-end performance continues to deteriorate.

Together with the increase in losses due to the absence of a route (D_{NR} ; Table II), these results support the observed mechanism: lower beacon freshness $\rightarrow$ route expirations $\rightarrow$ no-route events $\rightarrow$ DR degradation.

C. Indirect indicators of DC-induced postponement and contention (CAD)

Since this campaign does not report per-event waiting times in seconds, we use normalized counts as indirect indicators of DC and contention effects. Fig. 6(a) shows B_d , the number of DC blocking/postponement events per node, and Fig. 6(b) shows B_c , the number of busy CAD events per node (an indirect indicator of channel occupancy/contention; it does not equal

collision rate). Both increase with network size and load in the range where the network still attempts transmissions; however, at very high densities B_d may stabilize or even decrease because the system enters a contention-limited regime (low R_{tx0}), reducing the effective number of TX attempts that could be postponed by DC. In that regime, the dominant indicator becomes B_c , which continues to capture channel occupancy.

D. Latency: conditioning and sample size

Fig. 7 reports the mean end-to-end latency and the 95th percentile D_{95} as a function of N under low/medium loads and DC off/on (1%). Both the mean and D_{95} are computed *only over delivered packets*, which introduces survivorship bias when the delivered fraction is low: in scenarios with low DR and/or low R_{tx0} (fraction of generated packets that obtain at least one TX attempt at the source), the latency sample is dominated by “easy” packets that manage to be delivered.

This explains patterns that may look counterintuitive at first sight. In some *low* load cases with DC off, the mean latency and D_{95} are higher than with DC on. This does not mean that enabling DC improves the forwarding dynamics; instead, DC off admits and delivers a larger fraction of packets, including a non-negligible tail of packets that survive long in-network waits (contention, backoff and queueing at relays). Those “slow survivors” enter the latency statistics and can push D_{95} to higher values.

Conversely, under *medium* load we observe that DC off and DC on can exhibit similar D_{95} (and, to a lesser extent, mean latency), even though DR and R_{tx0} differ significantly. In this regime the system is already strongly contention-limited: many “difficult” packets fail to be delivered under either configuration, so the latency distribution of delivered packets is dominated by a subset of relatively “fast” packets. In such cases, D_{95} may remain almost constant while overall performance continues to degrade.

V. LIMITATIONS AND VALIDITY

Latency metrics (e.g., D_{95}) are reported conditioned on delivered packets and should be interpreted jointly with DR , R_{tx0} and the sample size N_{del} to avoid survivorship bias. Finally, B_d (DC blocking/postponement events per node) and B_c (busy CAD events per node) are used as relative indirect indicators of temporal restrictions and contention; they do not represent waiting times in seconds nor direct collision rates.

VI. CONCLUSIONS

This work reproducibly quantifies the impact of a 1% DC in multi-hop LoRa mesh networks with proactive DV routing and a CAD-based CSMA/CA MAC, implemented in ns-3. In a campaign of 240 runs with $N \in \{9, 16, 25, 36, 49, 64\}$, *low/medium* loads and 10 seeds per configuration, enabling DC is observed to systematically reduce the end-to-end delivery ratio and to degrade control-plane freshness: the fraction of effectively transmitted beacons (H_b) decreases, DV expirations (E_{DV}) emerge, and *no-route* events increase, contributing to E2E losses.

As N increases, the fraction of packets that reach at least one transmission attempt at the source (R_{tx0}) shows a transition towards a *contention-limited* regime, in which channel admission becomes the dominant limitation and mitigates the relative impact of DC on DR . The combination of metrics DR , R_{tx0} , H_b , E_{DV} , D_{NR} , B_d and B_c makes it possible to quantitatively isolate the contributions of contention and of temporal constraints due to DC, providing traceability of the mechanism behind the observed degradation.

As future work, sensitivity to alternative interference/capture models will be evaluated and the comparison will be extended to additional DV/MAC baselines, in addition to validation in LoRa testbeds. Likewise, we plan to report detailed lifetime and fairness metrics across the different scenarios and loads.

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